Type MS · Show how an image forms on the retina and reaches the brain

Answer 1 of 7 · Question 8 part a.i · 5 marks

2021 NOV · Paper 01 · extract 1 of 2 · find part a.i in this mark scheme extract

8	(a) (i)	• arrow heads showing direction of light rays; • bent at cornea; • bent at lens; • cross over in eyeball; • inverted image on retina;		5 marks
	(ii)	transmits/carries impulses from the retina/rod and cones; to the brain;		2 marks
	(b) (i)	behind the retina/fovea;		1 mark
	(ii)	eyeball too short / eye lens not powerful enough/too flat/not convex enough;		1 mark
	(iii)	use a convex / converging lens; to converge/refract/bend rays of light;		2 mark
	(c)	• infection; • rejection;		2 marks

Total 13 marks

Type MS · Show how an image forms on the retina and reaches the brain

Answer 1 of 7 · Question 8 part a.i · 5 marks

2021 NOV · Paper 01 · extract 2 of 2 · find part a.i in this mark scheme extract

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Type MS · Show how an image forms on the retina and reaches the brain

Answer 2 of 7 · Question 1 part b · 3 marks

2019 JUN · Paper 02 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

1(a)	line labelled A pointing to retina; line labelled B pointing to ciliary muscles; line labelled C pointing to iris; line labelled D pointing to optic nerve; line labelled E pointing to eyelid;		5

Question Number	Answer	Additional guidance	Mark
1(b)	bending at cornea; bending at lens; image formed on retina;		**3**

Question Number	Answer	Additional guidance	Mark
1(c)	long sight/myopia; short sight/hypermetropia; astigmatism;		3

Total 11 marks

Type MS · Show how an image forms on the retina and reaches the brain

Answer 2 of 7 · Question 1 part b · 3 marks

2019 JUN · Paper 02 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

Paper 2: Subject Human Biology

Question Number	Answer	Additional guidance	Mark

Type MS · Show how an image forms on the retina and reaches the brain

Answer 3 of 7 · Question 6 part a.ii · 4 marks

2025 JUN · Paper 02 · extract 1 of 2 · find part a.ii in this mark scheme extract

6	(a)	(i)	• A=rods/cones/retina; • detects light; • converts light energy into electrical energy; • (changed into (nerve) impulses; • B = transmits (nerve) impulses/electrical signals; • from (retina) to brain;	At least 1 mark from A and B to get full marks	4
		(ii)	• bent inwards at cornea/lens; • rays pass through the cornea and converge after the lens; • rays cross in eye (after lens); • inverted image/arrow;	Do not accept bending outwards for MP1 and MP2 Mp3- do not accept rays crossing before the retina	4
	(b)		• use both eyes and one eye; • use of eye patch/eq; • ball thrown same distance/stated distance;; • ball thrown many times; • count how many times ball is caught; • same type/size of ball;		4
	(c)		• misshapen/eq. cornea/lens; • light not focussed correctly/ light scatters/light not refracted properly; • blurred image;		2
					Total =14

Type MS · Show how an image forms on the retina and reaches the brain

Answer 3 of 7 · Question 6 part a.ii · 4 marks

2025 JUN · Paper 02 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Show how an image forms on the retina and reaches the brain

Answer 4 of 7 · Question 3 part c.i · 2 marks

2020 JAN · Paper 02 · find part c.i in this mark scheme extract

number				
3 (a)	(i)	**The only correct answer is C** lens ***A** is incorrect because this is where the image is formed* ***B** is incorrect because that is what controls the size of the pupil* ***D** is incorrect because the pupil is the hole in the middle of the iris*		1
	(ii)	• retina(1) • pupil (1) • choroid(1)		3
(b)		Value between 190 and 230 (μm)		1
(c)	(i)	• transmit (nerve/electrical) impulses(1) • to the brain/CNS/relay neurones(1)		2
	(ii)	• less/no impulses able to reach the brain(1) • vision blurred/less clear/less focussed/image not formed/seen/reference to blindness(1)	Allow signals	2

Total for Question 3 = 9 marks

Type MS · Show how an image forms on the retina and reaches the brain

Answer 5 of 7 · Question 3 part c.ii · 2 marks

2020 JAN · Paper 02 · find part c.ii in this mark scheme extract

Question number					
3	(a)	(i)	**The only correct answer is C** lens ***A** is incorrect because this is where the image is formed* ***B** is incorrect because that is what controls the size of the pupil* ***D** is incorrect because the pupil is the hole in the middle of the iris*		1
		(ii)	• retina(1) • pupil (1) • choroid(1)		3
	(b)		Value between 190 and 230 (μm)		1
	(c)	(i)	• transmit (nerve/electrical) impulses(1) • to the brain/CNS/relay neurones(1)		2
		(ii)	• less/no impulses able to reach the brain(1) • vision blurred/less clear/less focussed/image not formed/seen/reference to blindness(1)	Allow signals	2

Total for Question 3 = 9 marks

Type MS · Show how an image forms on the retina and reaches the brain

Answer 6 of 7 · Question 8 part a.ii · 2 marks

2021 NOV · Paper 01 · extract 1 of 2 · find part a.ii in this mark scheme extract

8 (a)	(i)	• arrow heads showing direction of light rays; • bent at cornea; • bent at lens; • cross over in eyeball; • inverted image on retina;		5 marks
	(ii)	transmits/carries impulses from the retina/rod and cones; to the brain;		2 marks
(b)	(i)	behind the retina/fovea;		1 mark
	(ii)	eyeball too short / eye lens not powerful enough/too flat/not convex enough;		1 mark
	(iii)	use a convex / converging lens; to converge/refract/bend rays of light;		2 mark
(c)		• infection; • rejection;		2 marks

Total 13 marks

Type MS · Show how an image forms on the retina and reaches the brain

Answer 6 of 7 · Question 8 part a.ii · 2 marks

2021 NOV · Paper 01 · extract 2 of 2 · find part a.ii in this mark scheme extract

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Type MS · Show how an image forms on the retina and reaches the brain

Answer 7 of 7 · Question 3 part a.ii · 2 marks

2024 NOV · Paper 01 · extract 1 of 2 · find part a.ii in this mark scheme extract

3 (a)	(i)	Part W - pupil; Part X – retina;	1 mark for each correctly named part	2
	(ii)	• sends (electrical) impulses from photoreceptors/rods/cones; • to the brain/CNS; OR • sends (electrical) impulses from brain/CNS; • to iris/ciliary muscles;		2
	(iii)	lens becomes thinner/flatter/less convex; less refraction;	allow less light refracted	2
	(iv)	Any two from: • shape of cornea restored; • light refracted, onto the retina/correctly/notmally; • correct information sent to the brain; • image/vision not blurred/clear;		2
(b)		• measure the size/diameter of the pupil; • in a dark room; • shine a light into the eye; • measure/compare size/diameter of the pupil;	accept ref to different light intensities	4

Total Marks 12

Type MS · Show how an image forms on the retina and reaches the brain

Answer 7 of 7 · Question 3 part a.ii · 2 marks

2024 NOV · Paper 01 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks